1 A sample of material has cross-sectional area A and length L. The temperatures at the two sides of the sample are T_1 and T_2. Thermal energy Q is transferred through the sample in time t.

These quantities are related by

$$\frac{Q}{t} = \frac{k \times A \times (T_1 - T_2)}{L}$$

What are the SI base units of k?

A kg m s^{-3} $^{o}C^{-1}$ **B** kg m s^{-3} K^{-1} **C** kg m s^{-1} $^{o}C^{-1}$ **D** kg m s^{-1} K^{-1}